

Deep Learning & Neural Networks

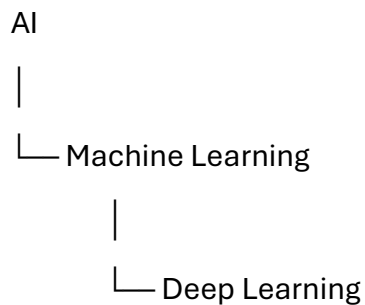
✓ CNN (Convolutional Neural Networks)

✓ RNN (Recurrent Neural Networks)

Topic 1: What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple hidden layers.

Hierarchy



Real Examples

- ChatGPT
- Copilot
- Siri
- Face Recognition
- Self Driving Cars
- Voice Assistants

Topic 2: What is a Neural Network?

A Neural Network is inspired by the human brain.

It consists of:

Input Layer

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Hidden Layers

↓

Output Layer

Example

Employee Age

Experience

Salary

↓

Neural Network

↓

Will Employee Leave?

Topic 3: Neural Network Architecture

Input Layer



Hidden Layer 1



Hidden Layer 2



Output Layer

Components

Neuron

Small processing unit.

Weight

Importance assigned to input.

Bias

Adjusts prediction.

Activation Function

Determines whether the neuron should activate.

Examples:

- ReLU
- Sigmoid
- Softmax

Topic 4: CNN (Convolutional Neural Networks)

CNN is mainly used for images.

Examples

- Face Recognition
- Medical Imaging
- Object Detection
- Security Cameras

CNN Architecture

Image

↓

Convolution Layer

↓

Pooling Layer

↓

Fully Connected Layer

↓

Prediction

Real Example

Image

↓

CNN

↓

Cat = 99%

Dog = 1%

Topic 5: RNN (Recurrent Neural Networks)

RNN is designed for sequence data.

Examples

- Language Translation
- Speech Recognition
- Time Series Forecasting

RNN Architecture

Word1

↓

Word2

↓

Word3

↓

Prediction

Example

I love Azure

RNN understands word sequence.

Topic 6: Transformers

Transformers are the foundation of modern Generative AI.

Used by:

GPT

ChatGPT

Copilot

Gemini

Claude

Llama

Why Transformers?

Older models:

RNN

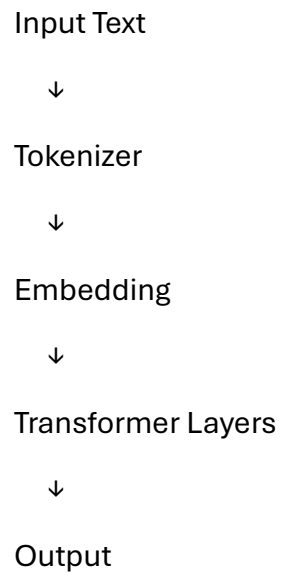
Problems:

- Slow
- Difficulty remembering long context

Transformers solve:

- ✓ Long Context
 - ✓ Faster Training
 - ✓ Better Accuracy
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Transformer Architecture



Example

Prompt:

Explain Azure Service Bus

Transformer:

Understands Context



Generates Response

Topic 7: Why ChatGPT Works

ChatGPT uses:

Transformer Architecture

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Large Language Model

↓

Generative AI

ChatGPT Simplified Architecture

User Prompt

↓

Tokenizer

↓

Embeddings

↓

Transformer

↓

Generated Response

Topic 8: Deep Learning Use Cases

Vision

- Face Recognition
- OCR
- Medical Imaging

Audio

- Speech Recognition
- Voice Generation

NLP

- ChatGPT
- Copilot
- Translation

Autonomous Systems

- Self Driving Cars
- Robotics